WEAKLY ETALE MORPHISMS OF SCHEMES COMPOSE, VIA FLATNESS OF ITERATED DIAGONALS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18973-18982 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18984-18999, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. First card of the chapter's weakly etale trilogy in this wave; used by the next two cards in the same wave.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of weakly etale (flat plus flat diagonal, More on Morphisms definition section), stability of flat morphisms under base change and composition (Morphisms, lemmas base-change-flat and composition-flat), and the fibre-product identity for iterated diagonals.

Source reference. Original lemma 01U9: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18999–18999. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01U7: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18999–18999. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to Y \to Z$ be morphisms of schemes.*

(1) *If $X \to X \times_Y X$ and $Y \to Y \times_Z Y$ are flat, then $X \to X \times_Z X$ is flat.*
(2) *If $X \to Y$ and $Y \to Z$ are weakly étale, then $X \to Z$ is weakly étale.*

Proof. Part (1) follows from the factorization

$$X \to X \times_Y X \to X \times_Z X$$

of the diagonal of X over Z, the fact that

$$X \times_Y X = (X \times_Z X) \times_{(Y \times_Z Y)} Y,$$

the fact that a base change of a flat morphism is flat, and the fact that the composition of flat morphisms is flat (Morphisms, Lemmas 01U9 and 01U7). Part (2) follows from part (1) and the fact (just used) that the composition of flat morphisms is flat. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.